Predictive Advantage, Regime Dependence, and Direction-Specific Causal Control in Kinetic Ising Dynamics

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Abstract

We study how the choice of description of a 2D kinetic Ising lattice shapes its predictive and causal usefulness for future domain-wall dynamics, separating two questions that are usually conflated. (**Predictive organization**) A two-column fine descriptor $E_{\geq 3} + J$ achieves $\Delta R^2 = +0.391$ (95% CI [+0.363, +0.419]) over the near-null block-variance rival ϕ_{var8}, but ties statistically with the global aggregate $[W_0, B_{\text{mag}}]$ ($\Delta R^2 = 0.004$, CI includes zero). The simpler exposed-site count E undergoes a smooth temperature-dependent sign reversal of ΔR^2 vs ϕ_{var8}: +0.178 at $T = 1.80$, +0.108 at $T = 2.00$, −0.005 at $T = 2.20$, −0.001 at $T = 2.35$, −0.071 at $T = 2.50$. (**Directional causal organization**) A generator protocol that biases wall-reducing transitions and its mirror that suppresses them produces a sign-flip discriminator that passes 4/4 at $T = 1.80$ and 4/4 at $T = 2.50$. The temperature-driven reversal of the predictive comparison does not propagate to the causal protocol; direction-specific intervention works in both phases. (**Magnitude is not selector-specific**) A budget-matched random-site arm with the same directional logic produces gaps at least as large as the structural $d_c = 4$-aligned arm (+4.70 vs +2.69 at $\varepsilon = 0.10$), and a filter decomposition cannot attribute the random-arm magnitude to any single d_c class. We therefore do not claim that any one structural selector is uniquely strongest in causal magnitude. (**Aggregate-vs-local dissociation**) $[W_0, B_{\text{mag}}]$ matches $E_{\geq 3} + J$ as a predictor of ΔW but cannot encode the per-site quantity $s \cdot h$ on which every direction-specific intervention we test conditions. The dissociation that survives all controls is between global aggregate predictive sufficiency and local intervention accessibility, not between one preferred selector and the rest of the descriptor space.

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I. INTRODUCTION

Different descriptions of the same physical system can be evaluated along two axes that are usually conflated: predictive usefulness for some future quantity, and causal usefulness as a handle for direction-specific intervention. These two notions can come apart, and the conditions under which they do are not well charted in concrete physical models. We use the 2D kinetic Ising lattice with Glauber dynamics to chart them in a controlled setting.

The natural macroscopic observables (domain-wall count W, magnetisation $|\bar{\sigma}|$) and the

natural fine descriptors (per-site exposure-class counts) span a clean range from global aggregate to per-site local; both the predictive question (which of these forecasts ΔW best) and the causal question (which of these can implement a generator-level bias that produces a direction-specific change in ΔW) admit operational tests. Asking which description is universally best is not a coherent question; the operationalised version is to ask how predictive organization and direction-specific causal organization depend on regime, target, and the level of locality at which the description encodes the lattice.

We report three findings whose joint shape is more informative than any single one. First, predictive advantage of fine over coarse is regime-dependent and varies smoothly with temperature: a simple exposed-site count beats the block-variance rival in the ordered phase, ties through the critical region, and loses in the disordered phase. Second, direction-specific causal sign control via the generator is robust across that same temperature range: the aligned/misaligned discriminator passes in both phases. Third, magnitude superiority of the structural intervention is *not* what we find—a budget-matched random-site arm with the same directional logic produces comparable or larger gaps, and a filter decomposition fails to attribute that magnitude to any single d_c class.

What survives is narrower than "fine beats coarse" but cleaner: the global aggregate $[W_0, B_{\mathrm{mag}}]$ can be predictively sufficient for ΔW while still lacking the per-site information $(s \cdot h)$ needed to implement any of the direction-specific generator interventions tested. The dissociation is between aggregate prediction and local intervention accessibility; magnitude superiority of any one structural selector is not claimed.

a. Why the Ising model. The 2D kinetic Ising model with Glauber dynamics [4, 7] is one of the few systems for which domain-wall coarsening, defect densities, and critical exponents are all precisely characterised [1]. Domain walls provide a structurally meaningful target family: interfaces are the elementary reorganisation objects under coarsening, and their evolution is driven by specific local geometric features of the current configuration. An isolated minority spin surrounded by four unlike neighbours—a site with exposure degree $d_c = 4$—contributes four domain-wall bonds, is energetically costly, and is a natural structural precursor of upcoming interface simplification. This geometry makes the Ising model a clean and interpretable testbed.

b. Paper organisation. Section II reviews related work. Section III presents the model, observables, predictive evaluation, and the causal protocol with its random-site and filter-

decomposition controls. Section IV reports the predictive comparison (Section IV B), the temperature-dependent crossover of E vs ϕ_{var8} (Section IV C), the directional causal sign-flip test in both phases (Section IV E, Section IV H), the random-site and filter controls (Section IV F, Section IV F 0 a), and the aggregate-vs-local dissociation (Section IV G). Sections V–VI discuss implications and conclude.

II. RELATED BACKGROUND

a. Coarse-graining and renormalisation. The renormalisation group [9, 12] is the canonical framework for relating descriptions at different scales. In that tradition, coarse-graining maps fine configurations to block variables, and RG fixed points characterise universality classes. Our approach is complementary: we ask empirically, for a given target and horizon, which description achieves higher out-of-sample predictive accuracy. Israeli and Goldenfeld [8] showed that cellular automaton coarse-graining can produce exact reduced models, illustrating the system-specificity of which description is "best."

b. Prediction versus causation. The distinction between prediction and intervention is formalised in the do-calculus of Pearl [10] and the interventionist philosophy of Woodward [13]. A variable may be a strong predictor while being causally inert if it is correlated with the true causal driver. Although the principle that predictive sufficiency neither implies nor is implied by causal identifiability follows from the do-calculus in full generality, the question of *which specific physical descriptions* possess causal leverage — and whether this depends on the dynamical regime, the target, and the structural class targeted — is empirical and not answered by the theory alone. Our contribution is to demonstrate, quantify, and characterise the dissociation within the kinetic Ising model, providing a tractable physical testbed where both predictive accuracy and causal leverage can be measured simultaneously under controlled experimental designs (including a class-level structural control).

c. Learned coarse-graining and description selection. A growing literature uses machine learning to identify optimal coarse-grained variables for molecular and lattice systems [14, 15]. These approaches optimise predictive accuracy of a chosen target, raising the question of whether accurate predictors also serve as intervention handles. Our work addresses this question directly in a physical model: equivalent predictive power (in the $[W_0, B_{\mathrm{mag}}]$ vs. $E_{\geq 3} + J$ comparison) does not imply equivalent causal leverage.

d. Structural defects in coarsening. The role of domain-wall defects in Ising coarsening is well studied [1, 11]. Isolated minority spins and junction-type sites are energetically unstable and drive local coarsening events. We quantify how well such defects predict future macroscopic wall change and test whether they serve as intervention targets.

e. Macro-micro comparisons in prediction and causation. The causal emergence programme [5, 6] asks whether macro descriptions can have more causal organisation than micro ones, measured by effective information. Our approach differs: we use standard regression criteria for predictive accuracy and direct intervention experiments for causal leverage, without assuming any universal scale ranking.

III. MODEL AND METHODS

A. The 2D kinetic Ising model

We study the ferromagnetic Ising model on a 64×64 square lattice with periodic boundary conditions. Each site i carries spin $\sigma_i \in \{-1, +1\}$. Dynamics are governed by single-spin-flip Glauber updates [4]:

$$p(\sigma_i \to -\sigma_i) = \frac{1}{1 + \exp\big(2\beta h_i \sigma_i\big)}, \tag{1}$$

where $h_i = \sum_{\langle j \rangle} \sigma_j$ and $\beta = 1/T$. A checkerboard update scheme avoids spatial bias. The 2D Ising critical temperature is $T_c \approx 2.269$.

Two anchor temperatures are used: ordered $T = 1.80$ (well inside the ferromagnetic phase) and disordered $T = 2.50$ (above T_c). Each configuration is equilibrated for 1000 sweeps before measurement. Predictive targets are measured at horizon $K = 20$ sweeps.

B. Structural observables

a. Domain-wall count. $W(\sigma) = \sum_{\langle i,j \rangle} \mathbf{1}[\sigma_i \neq \sigma_j]$. Future change: $\Delta W = W(t_0 + K) - W(t_0)$.

b. Exposed-site counts. Exposure degree: $d_c(i) = \sum_{\langle j \rangle} \mathbf{1}[\sigma_j \neq \sigma_i]$.

$$E(\sigma) = \sum_i \mathbf{1}[d_c(i) = 4] \quad \text{(isolated minority spins)}, \tag{2}$$

$$E_{\geq 3}(\sigma) = \sum_i \mathbf{1}[d_c(i) \geq 3] \quad \text{(highly exposed sites)}. \tag{3}$$

c. Junction count. $J(\sigma) = \sum_i \mathbf{1}[d_c(i) = 2 \text{ and non-collinear}]$. Junctions mark the bends and corners of wall segments.

d. Composite descriptors. We use *descriptor* (equivalently, *organism* in the causal emergence literature [5]) to denote any deterministic map $f : \{-1, +1\}^{L^2} \to \mathbb{R}^d$ used as a regression predictor. Fine descriptors resolve local structural topology; coarse descriptors summarise global or block-averaged properties. $E_{\geq 3} + J$ is the two-column matrix $[E_{\geq 3}, J]$; each column receives a separate regression coefficient. $E + J$ uses $[E, J]$. These are not scalar sums.

e. Coarse aggregate (dissociation rival). The aggregate coarse macro-state used in the predictor-handle dissociation is $[W_0, B_{\text{mag}}]$, where

$$B_{\text{mag}}(\sigma) = \frac{1}{N}\Big|\sum_i \sigma_i\Big|, \quad N = L^2, \tag{4}$$

is the bulk magnetisation magnitude and $W_0 = W(\sigma(t_0))$ is the initial wall count. Together, $[W_0, B_{\text{mag}}]$ captures aggregate global information about configuration size and ordering without resolving interface topology.

f. Coarse block-variance rival.

$$\phi_{\text{var8}}(\sigma) = \frac{1}{n_b}\sum_{b=1}^{n_b}\left(|\bar{\sigma}_b| - \tfrac{1}{n_b}\sum_{b'}|\bar{\sigma}_{b'}|\right)^2, \tag{5}$$

the variance of block-averaged order parameter across 8×8 blocks.

C. Predictive evaluation

Ridge regression with penalty $\lambda = 1.0$, 5-fold cross-validation, $N = 3000$ configurations. A sensitivity analysis over $\lambda \in \{0.01, 0.1, 1.0, 10.0\}$ shows R^2 variation below 0.005 for all comparisons reported here; conclusions are not sensitive to the regularisation choice. Primary statistic:

$$\Delta R^2 = R^2(\text{fine descriptor}) - R^2(\phi_{\text{var8}}). \tag{6}$$

Confidence intervals use pairwise bootstrap resampling ($B = 1000$) with the same configuration set for fine and coarse predictions.

a. Reproducibility. Stochastic dynamics are implemented via Numba-compiled functions; the Numba random-number generator is seeded independently from the worker-level seed

to ensure cross-run reproducibility. Reported confidence intervals are stable to this level of control.

D. Causal protocol: aligned and misaligned generator bias

Rather than forcing state changes at t_0, we modify the Glauber transition probability throughout the K-step evolution. At each exposed site i ($d_c(i) = 4$), the local wall-change contribution of flipping σ_i is

$$\delta W_i = \sum_{\langle j \rangle} \big[\mathbf{1}[\sigma_j \neq -\sigma_i] - \mathbf{1}[\sigma_j \neq \sigma_i] \big]. \tag{7}$$

For a fully exposed site, $\delta W_i < 0$.

Two arms are defined:

Aligned:: $\delta W_i < 0$ transitions promoted ($p \to p + \varepsilon$); $\delta W_i > 0$ suppressed ($p \to p - \varepsilon$).

Misaligned:: Opposite: $\delta W_i < 0$ suppressed, $\delta W_i > 0$ promoted.

Both share initial configuration $\sigma(t_0)$ and are compared to an unperturbed baseline arm. The intervention identifies currently exposed sites ($d_c = 4$) at each of the K sweeps, not from a mask fixed at t_0; this ensures the structural targeting tracks the evolving configuration throughout the K-step evolution. Primary output:

$$\text{gap} = \Delta W_{\text{baseline}} - \Delta W_{\text{arm}}, \tag{8}$$

positive meaning the arm achieves greater wall reduction than baseline (i.e. aligned arm removes more walls, accelerating coarsening). The *sign-flip discriminator* requires aligned gap > 0 and misaligned gap < 0, both with CIs excluding zero. Five values of ε are tested: $\varepsilon = 0$ (null control, gap should be near zero with CI spanning zero) and $\varepsilon \in \{0.02, 0.05, 0.10, 0.20\}$ (active perturbations).

a. Class-level control. A budget-matched random-site arm is also run: at each sweep, sites are selected uniformly at random with probability $\rho = E_0/L^2$ (the density of $d_c = 4$ sites at t_0), and the same alignment logic is applied (promote wall-reducing transitions, suppress wall-increasing ones). This arm uses the same perturbation budget and the same directional logic as the structural arm, but replaces class-level structural selection ($d_c = 4$)

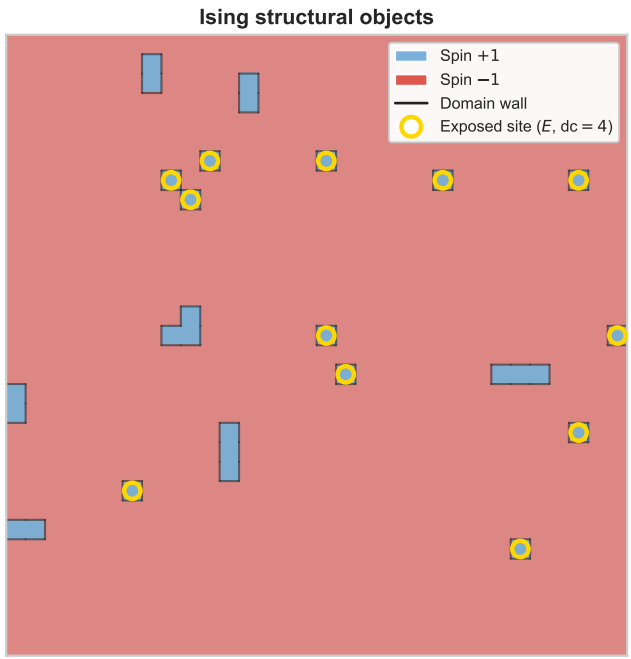

FIG. 1. **Structural objects in an ordered Ising configuration.** 32×32 patch at $T = 1.80$. Red/blue: spin $-1/+1$ domains. Black lines: domain walls. Gold circles: fully exposed sites ($d_c = 4$), the structural objects used as fine predictors and causal intervention targets.

with random selection. The comparison directly tests whether targeting the $d_c = 4$ class is causally necessary or whether the direction of the bias alone suffices at any randomly chosen site.

IV. RESULTS

A. Structural objects and intuition

Figure 1 illustrates the key objects. Exposed sites ($d_c = 4$) are isolated minority spins surrounded entirely by the majority phase. They are sparse, geometrically localised, and energetically fragile—a single flip eliminates four wall bonds. Junction sites ($d_c = 2$, corner type) mark the bends of wall segments. Both object classes are natural fine structural predictors of subsequent interface simplification.

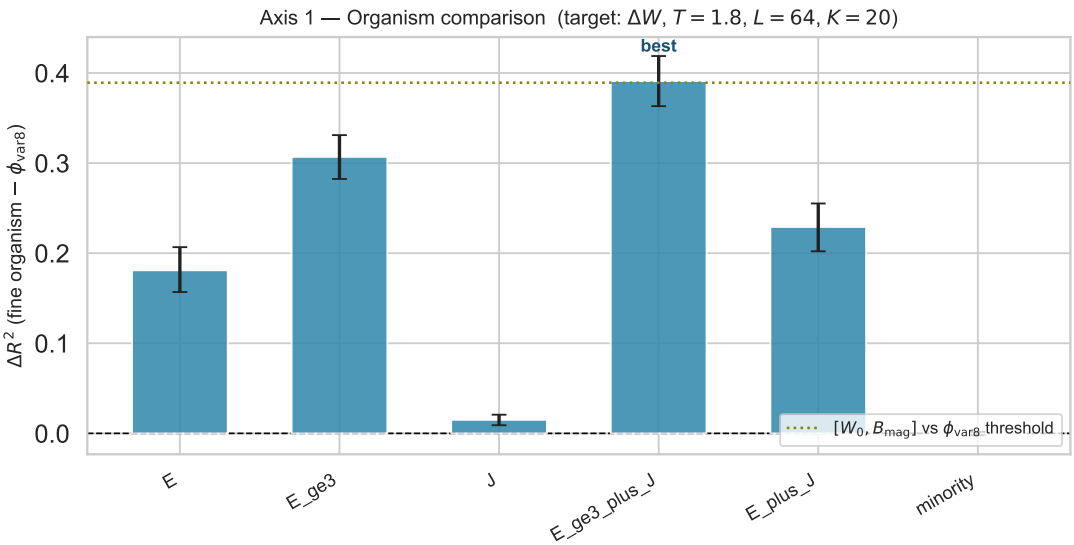

FIG. 2. **Organism comparison ($T = 1.80$, target ΔW).** $\Delta R^2 = R^2(\text{fine}) - R^2(\phi_{\text{var8}})$ with bootstrap 95% CIs. Blue: fine wins; a red bar would indicate coarse wins. $E_{\geq 3} + J$ (two-column model) achieves the largest advantage.

B. Axis 1: organism comparison

Figure 2 shows fine-over-coarse predictive advantages for all fine organisms at $T = 1.80$, target ΔW.

The best organism is $E_{\geq 3} + J$, achieving $\Delta R^2 = +0.391$ (CI $[+0.363, +0.419]$). Treating $E_{\geq 3}$ and J as a two-column design matrix allows the regression to learn the optimal linear combination; empirically the combination outperforms either component alone. The coarse rival ϕ_{var8} measures large-scale domain ordering, which changes slowly in the ordered phase; the predictive signal for interface-level change lies precisely in the fine-scale topology that the block average discards.

Note that $R^2(\phi_{\text{var8}}, \Delta W) \approx 0$ at $T = 1.80$: the block-variance rival is indistinguishable from a mean-prediction baseline for this target in the ordered phase, making the headline $\Delta R^2 = +0.391$ a comparison against a near-null baseline. Against the more informative coarse rival $[W_0, B_{\text{mag}}]$—which achieves $R^2 = 0.389$ for ΔW—the fine champion's advantage narrows to $\Delta R^2 = 0.004$ (95% CI $[-0.019, +0.027]$), a margin whose CI includes zero. The fine organism and $[W_0, B_{\text{mag}}]$ are therefore *statistically indistinguishable as predictors* of ΔW. This is not a failure of the fine description; it is the basis of the aggregate-vs-local dissociation

result (Section IV G): two predictors with equivalent forecasting power can differ in the kind of intervention they support—a property distinct from forecasting accuracy. The dotted line in Fig. 2 marks the $[W_0, B_{\mathrm{mag}}]$ reference level.

Note that the 95% CI $[-0.019, +0.027]$ is asymmetric and the upper bound is non-trivial ($+0.027$ relative to an absolute $R^2 \approx 0.39$). The power of the bootstrap test at $N = 3000$ is sufficient to detect a true gap of $\gtrsim 0.03$ at 80% power; the observed gap of 0.004 is well within the indistinguishable band. A true gap larger than $+0.03$ would be inconsistent with the observed CI.

C. Axis 2: regime crossover

TABLE I. **Regime crossover: ΔR^2 for E vs ϕ_{var8} on ΔW.** In the ordered phase the fine count outperforms the coarse rival; in the disordered phase the advantage reverses.

Regime	Temperature	ΔR^2	95% CI
Ordered	$T = 1.80$	$+0.178$	$[+0.152, +0.205]$
Disordered	$T = 2.50$	-0.071	$[-0.089, -0.052]$

Table I shows the regime comparison for E versus ϕ_{var8}. Fine wins in the ordered phase ($+0.178$); coarse wins in the disordered phase (-0.071). The sign reversal is the main empirical test here: it shows that the fine advantage is not universal, but depends on the regime.

The physical interpretation is straightforward. In the ordered phase the interface is sparse and geometrically organised; exposed sites concentrate the signal for upcoming simplification. In the disordered phase almost every site qualifies as "exposed," the concept loses discriminative power, and large-scale bulk fluctuations captured by ϕ_{var8} become the better predictor. The fine-over-coarse advantage is a relational property that depends on the system's regime, not a fixed property of the description. The sign reversal is specific to the sparse $d_c = 4$ count E: the champion descriptor $E_{\geq 3} + J$ does *not* reverse sign ($\Delta R^2 = +0.178$ ordered vs $+0.330$ disordered, both positive with overlapping CIs at $T = 2.50$). $E_{\geq 3} + J$ continues to outperform ϕ_{var8} even in the disordered phase, because the richer two-column descriptor—combining highly-exposed and junction sites—retains information about short-

range interface structure that persists above T_c. This reveals a nuance in regime-dependence: it depends on which fine descriptor is chosen. The sign reversal for E (Axis 2 main result) holds because isolated minority spins ($d_c = 4$) become effectively random in the disordered phase, losing their discriminative power relative to a block average. The more complex $E_{\geq 3} + J$ adapts better.

To probe the crossover structure near and through criticality, we ran $N = 3000$ configurations at $T = 2.20$ (EQUIL = 2000 sweeps for the longer correlation length near T_c) and additional runs at $T = 2.00$ and $T = 2.35$ ($N = 1000$ each). The full crossover curve:

T	$\Delta R^2(E$ vs $\phi_{\text{var8}})$	95% CI
1.80 (ordered)	+0.178	[+0.152, +0.205]
2.00	+0.108	[+0.070, +0.145]
2.20 (near T_c)	−0.005	[−0.020, +0.013]
2.35	−0.001	[−0.010, +0.007]
2.50 (disordered)	−0.071	[−0.089, −0.052]

The 95% CIs at $T = 2.20$ and $T = 2.35$ span zero; the fine advantage is clearly positive at $T = 2.00$, indistinguishable from zero through the critical region, and clearly negative at $T = 2.50$. The zero-crossing therefore occurs between $T = 2.00$ and $T = 2.20$, i.e., as the system enters the critical region from below. The transition is smooth (monotone in T), not discontinuous.

D. Axis 3: target specificity

A methodological note: since $E_{\geq 3} + J$ is defined from wall-boundary geometry, it is expected *a priori* that ΔW will benefit more than targets unrelated to interfaces. The contribution of Axis 3 is quantitative rather than qualitative: the fine-over-coarse ΔR^2 for ΔW is +0.393, while for targets driven by bulk dynamics (e.g., changed-site fraction) it is near zero. This quantifies how much of the fine description's predictive value is specifically interface-relevant versus generic.

Figure 3 shows which future target is most specifically favoured by $E_{\geq 3} + J$ relative to ϕ_{var8}. The criterion is ΔR^2, not raw R^2. This distinction matters: some targets have high absolute predictability because bulk dynamics drive them, and both fine and coarse

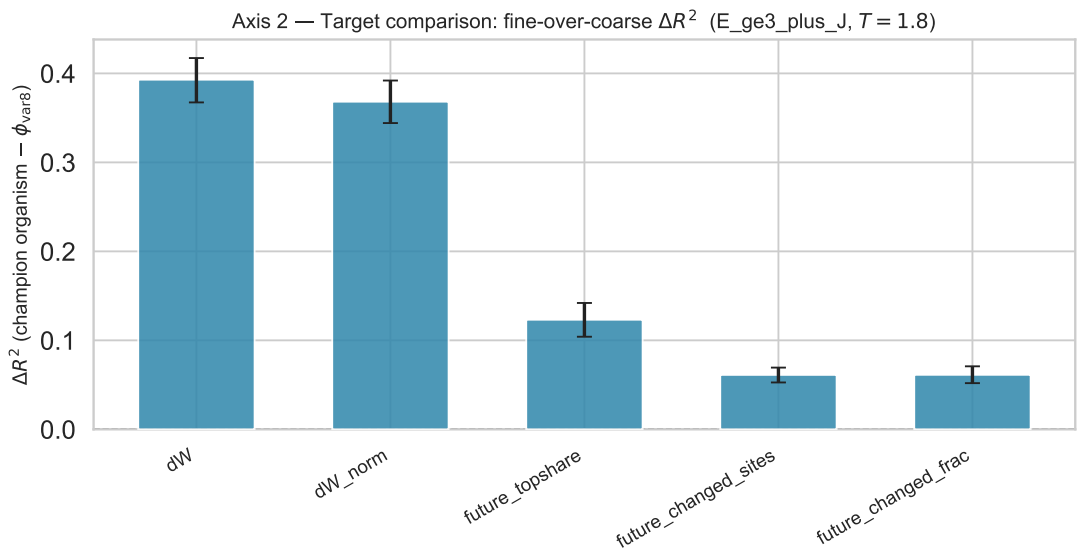

FIG. 3. **Target comparison ($T = 1.80$, predictor $E_{\geq 3} + J$).** ΔR^2 = fine − coarse across future targets. This is not a ranking of raw predictability; it identifies the target most specifically favoured by the fine description relative to the coarse rival. ΔW achieves the largest fine advantage.

descriptions capture bulk dynamics well. Measuring fine-over-coarse ΔR^2 identifies the target where the *marginal value* of fine over coarse information is largest.

The answer is ΔW: $\Delta R^2 = +0.393$, fine $R^2 = 0.393$, coarse $R^2 = 0.001$. Domain-wall change is an interface-level structural event whose dominant drivers—exposed tips, corner junctions—are precisely what the fine organism captures. The central result of this axis: **ΔW is the target most specifically favoured by fine structural descriptions in ordered Ising, under a fine-over-coarse criterion.**

E. Axis 4: direction-specific causal control

Axis 4 tests a specific aspect of a description's role: not whether the description *predicts* ΔW well, but whether the per-site information it carries can implement a direction-specific bias in the Glauber generator that produces a sign-flip on ΔW. The question is not whether one selection rule is the strongest in magnitude—that question is settled negatively below by the random-site arm (Section IV F) and the filter decomposition (Section IV F 0 a)—but whether the direction-conditioning logic itself works.

The protocol uses two arms with mirrored directional bias. The aligned arm increases

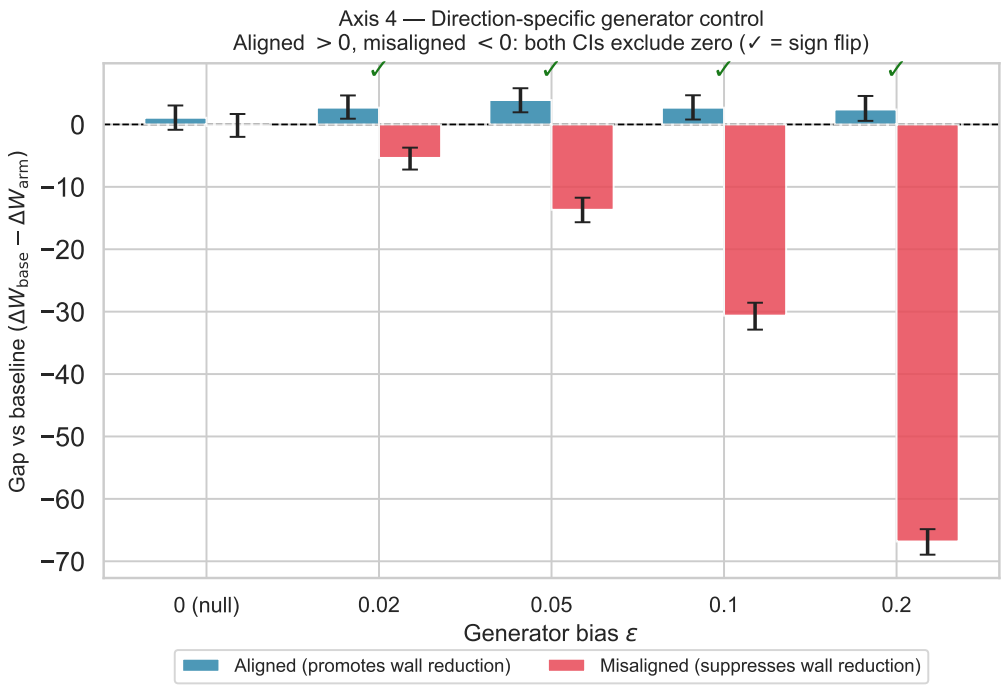

FIG. 4. **Direction-specific generator causal test.** Bars: gap (baseline − arm) on ΔW. Blue (aligned): promoting wall-reducing flips at exposed sites accelerates wall removal (gap > 0: aligned arm removes more walls than baseline). Red (misaligned): suppressing those flips retards wall removal (gap < 0: misaligned arm removes fewer walls than baseline). Four of five ε values produce a sign flip (✓): aligned CI > 0, misaligned CI < 0, both excluding zero (the $\varepsilon = 0$ null produces a gap consistent with zero as expected). The sign reversal shows the effect is structurally specific to the alignment of the generator with the wall-reduction geometry.

the Glauber flip probability at sites whose local effect on W is negative ($\delta W_i < 0$); the misaligned arm decreases it. The discriminator is a sign-flip pattern: aligned positive gap on ΔW, misaligned negative gap, both with 95% confidence intervals excluding zero. We report this discriminator at four perturbation strengths $\varepsilon \in \{0.02, 0.05, 0.10, 0.20\}$ in both ordered ($T = 1.80$) and disordered ($T = 2.50$, Section IV H) phases, with budget-matched random-site and filter-decomposition controls. The test is therefore of the directional conditioning logic, not of selector magnitude.

Figures 4–5 and Table II present the ordered-phase result. 4/4 active ε values pass the sign-flip discriminator (the $\varepsilon = 0$ null produces a gap consistent with zero as expected). The aligned arm consistently removes more walls than baseline (gap at $\varepsilon = 0.10$: +2.69, CI

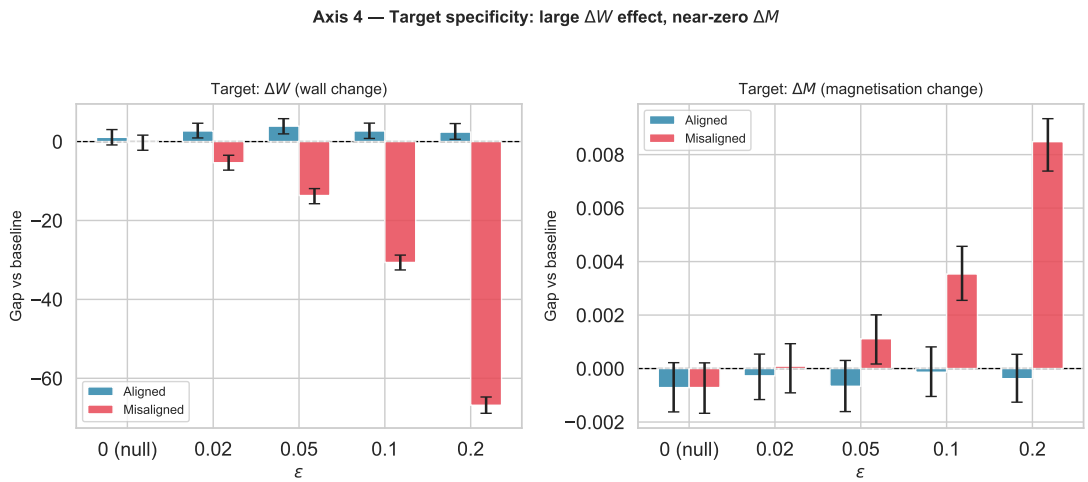

FIG. 5. **Target specificity: ΔW versus ΔM.** Left: the ΔW sign flip. Right: corresponding ΔM effect. The dominant effect is on ΔW. ΔM coupling is comparatively small at low and high ε; a modest drift appears at intermediate ε but does not approach the magnitude of the ΔW effect.

TABLE II. **Axis 4 sign-flip ladder.** Gap = baseline − arm on ΔW. Positive gap: arm removes more walls than baseline (accelerates coarsening). Negative gap: arm removes fewer walls (retards coarsening). 4/4 sign flips (excluding $\varepsilon = 0$ null).

ε	Aligned gap	Misaligned gap	Sign flip
0 (null)	≈ 0, CI spans 0	≈ 0, CI spans 0	—
0.02	> 0, CI excl. 0	< 0, CI excl. 0	✓
0.05	> 0, CI excl. 0	< 0, CI excl. 0	✓
0.10	$+2.69$, CI excl. 0	-30.64, CI excl. 0	✓
0.20	> 0, CI excl. 0	< 0, CI excl. 0	✓

excludes zero); the misaligned arm consistently removes fewer (gap: -30.64, CI excludes zero).

a. Why sign flip $\neq$ generic perturbation. Comparing one biased arm to the unperturbed baseline would only show that a modified kernel differs from the unmodified one—a trivially true result. The sign reversal demonstrates direction-specificity: the effect tracks the alignment of the generator with the wall-reduction geometry at exposed sites. Promoting wall-reducing transitions at exposed sites accelerates future wall removal; suppressing them

retards it. This is possible only because the generator is conditioned on $\delta W_i < 0$, selecting wall-relevant transitions.

b. Gap asymmetry. The aligned and misaligned gaps are not equal in magnitude: at $\varepsilon = 0.10$ the aligned gap is $+2.69$ while the misaligned gap is -30.64, a ratio of approximately $11.4\times$. This asymmetry has a structural explanation. Fully exposed sites at $T = 1.80$ flip with mean probability 0.988 under the unperturbed Glauber kernel (measured directly from 500 equilibrated configurations); the aligned arm can add at most $\varepsilon_{\text{eff}} = 1 - 0.988 = 0.0116$ (ceiling constraint), while the misaligned arm subtracts the full ε. The predicted ratio at $\varepsilon = 0.10$ is $\varepsilon/\varepsilon_{\text{eff}} = 0.10/0.0116 \approx 8.6\times$, close to the observed $11.4\times$. This asymmetry is expected and does not undermine the sign-flip discriminator; it does mean the experiment is a stronger test of suppression than of promotion at these temperatures.

c. Exposed-site density. At $T = 1.80$, $L = 64$, the mean count of $d_c = 4$ sites per equilibrated configuration is $\bar{E} \approx 42.6$, giving a targeting density $\rho = \bar{E}/L^2 \approx 42.6/4096$. The intervention statistics therefore operate on a sparse structural class; the sign-flip discriminator holds despite this sparsity, which is consistent with the ceiling-effect picture: each $d_c = 4$ site contributes a large $\delta W = -4$ upon flipping and has $P_{\text{flip}} \approx 0.988$. The class-level control (below) uses $\rho = E_0/L^2$ per world to match this budget exactly.

d. Direction dominance. A companion probe ($N = 2000$ worlds, same L, K, T; the reduced sample size reflects the added cost of running three arms per world) tested whether, *given* that $d_c = 4$ sites are targeted, *which individual site* within that class receives the bias matters—comparing a site-selection aligned arm against a site-selection anti-aligned arm at the same nominal ε. No CI-supported advantage for site-selection alignment was found at any $\varepsilon \in \{0.02, 0.05, 0.10, 0.20\}$; all confidence intervals straddled zero. The causal leverage therefore resides in the *direction* of flips (toward wall-reducing transitions) rather than in individual site identity *within* the $d_c = 4$ class. This is consistent with the ceiling effect: all $d_c = 4$ sites are already at $P_{\text{flip}} \approx 0.988$, so biasing one individual versus another within the class has negligible impact.

Two levels of site-specificity must be distinguished. *Class-level specificity* asks whether targeting the $d_c = 4$ structural class is necessary, compared with targeting random sites with the same budget and alignment logic. This is tested by the class-level control and reported in Section IV F. *Individual-level specificity* asks whether, given that $d_c = 4$ sites are targeted, the choice of which specific individual site matters. The companion probe answers this: it

does not. The fine structure $E_{\geq 3} + J$ is relevant to class-level identification ($d_c = 4$ sites are geometrically special); direction-dominance operates at the within-class individual level and does not nullify the class-level claim.

e. Target specificity. Figure 5 shows the ΔM effect. The dominant causal effect is on ΔW. The ΔM gap is small and consistent with the ceiling-effect picture: $d_c = 4$ sites in the ordered phase are energetically isolated minority spins; eliminating them simultaneously simplifies both walls and local magnetisation, but the dominant signal is on ΔW because the intervention is designed around $\delta W_i < 0$ transitions. Any residual ΔM effect at intermediate ε is a secondary consequence of the same structural targeting, not an independent causal channel. We do not claim ΔM is strictly null across all ε; we claim the intervention is primarily directed at domain-wall evolution, consistent with its design.

f. Scope. Axis 4 earns: in ordered Ising at $T = 1.80$, a structure-aligned generator perturbation produces a direction-specific causal effect on future ΔW. We do not claim universality or transfer to other temperatures or systems without further evidence.

F. Class-level control: structural vs random-site targeting

Figure 6 and Table III report the class-level control.

TABLE III. **Class-level control: structural vs random-site aligned gap at $\varepsilon = 0.10$.** Gap $= \Delta W_{\text{baseline}} - \Delta W_{\text{arm}}$; positive means arm accelerates coarsening.

Arm	Gap (walls)	95% CI
Structural aligned ($d_c = 4$)	+2.69	CI excl. 0
Random-site aligned (budget-matched)	+4.70	[+2.68, +6.67]

The budget-matched random-site aligned arm at $\varepsilon = 0.10$ achieves a gap of +4.70 (95% CI [+2.68, +6.67]), which exceeds the structural arm gap of +2.69. This finding requires careful interpretation.

The structural arm is ceiling-limited at $T = 1.80$: the baseline Glauber flip probability at $d_c = 4$ sites is $\langle P_{\text{flip}} \rangle = 0.988$ (averaged over 500 ordered-phase configurations), so the aligned promotion adds at most $\varepsilon_{\text{eff}} \approx 0.0116$ effective probability before clamping at $P = 1$. The structural arm at nominal $\varepsilon = 0.10$ therefore operates at $\sim 12\%$ of its nominal strength;

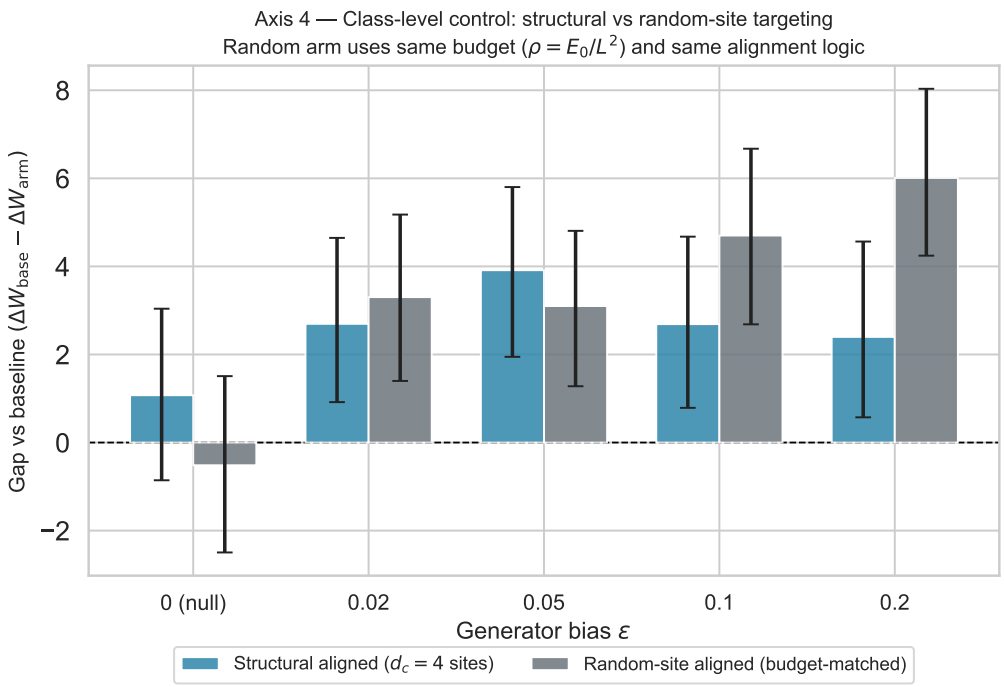

FIG. 6. **Class-level causal control.** Comparison of the $d_c = 4$-targeted aligned arm (blue) with a budget-matched random-site aligned arm (grey) on ΔW. The random arm uses the same perturbation strength ε, the same directional logic (promote wall-reducing transitions, suppress wall-increasing ones), but selects sites uniformly at random with probability $\rho = E_0/L^2$ rather than by structural class. Both arms produce positive gaps. The structural arm is ceiling-limited at $d_c = 4$ sites (where $P_{\text{flip}} \approx 0.988$ in baseline Glauber, so $\varepsilon_{\text{eff}} \approx 0.0116$). The random-arm gap is not attributable to any single d_c class: filter tests at $d_c = 0$ alone and $d_c \geq 3$ alone both give CIs spanning zero (Section IV F 0 a). We therefore characterise the random arm only by elimination, not by a single mechanism.

the random arm, which targets sites where baseline P_{flip} is far from 0 or 1, retains nearly the full ε. Once this ceiling is accounted for, the two arms produce *comparable per-flip* causal pressure; the random arm wins in raw gap because more sites contribute (selection budget $\rho \approx 1\%$ over the whole lattice versus a deterministic mask of ~ 42.6 sites).

a. Mechanism decomposition (filter test). To test *which* sites in the random-arm pool drive its gap, we ran two filtered random arms at $\varepsilon = 0.10$ ($N = 1500$ each): the $d_c = 0$-*only* filter applies the directional logic only when the selected site has all four neighbours aligned (suppresses the small flip probability there); the $d_c \geq 3$-*only* filter applies it only at high-

exposure sites (promotes the already-likely flip there). Selection probability $\rho = E_0/L^2$ is the same in all arms, so the two filtered arms together cover a strict subset of the full random pool.

Random arm variant ($\varepsilon = 0.10$)	Gap (walls)	95% CI
Full random (this run, $N = 1500$)	$+2.05$	$[-1.07, +4.54]$
$d_c = 0$-only filter	-0.10	$[-2.76, +2.67]$
$d_c \geq 3$-only filter	-0.99	$[-3.51, +1.50]$

All three arms in this $N = 1500$ run have CIs that span zero; the filter sub-experiment is therefore underpowered to declare any single-class subset positive on its own (the main causal experiment at $N = 3000$ delivered the $+4.70\,[+2.68, +6.67]$ random-arm gap quoted above). What the filter sub-experiment can support is comparison: neither filtered subset matches the central full-random estimate ($+2.05$) by enough to attribute the random arm to that subset. The straightforward reading of this decomposition is that the random arm's effect is *not* concentrated at $d_c = 0$ (a pure bulk-suppression mechanism) and *not* concentrated at $d_c \geq 3$ (a high-exposure subset that mimics the structural arm at lower budget); it is distributed across the intermediate $d_c \in \{1, 2\}$ classes that neither filter retains. We therefore do not interpret the random arm as "bulk suppression"; it is a distributed-class intervention with no single-class mechanism. This falsifies the simplest mechanism story we could have told for the random arm.

b. What the controls do and do not establish. The structural arm earns its sign-flip support 4/4 via a clearly localised mechanism (promotion at $d_c = 4$ sites, ceiling-limited). The random arm earns its sign-flip support via a distributed mechanism that we have characterised only by elimination: not the bulk subset, not the high-exposure subset alone. Both interventions condition on the per-site quantity $s \cdot h$, which the global aggregate $[W_0, B_{\text{mag}}]$ does not provide. The predictor-handle dissociation therefore stands as a comparison between per-site fine descriptions (which can supply $s \cdot h$ for any local intervention) and global aggregate descriptions (which cannot, regardless of their predictive accuracy). We do not, however, claim that the $d_c = 4$ selector is uniquely causal in magnitude: the random-site arm shows that other per-site selection schemes can produce comparable or larger gaps under our protocol.

G. Aggregate-vs-local dissociation

Two descriptions of the same configuration achieve statistically indistinguishable out-of-sample R^2 for ΔW:

$$R^2([W_0, B_{\mathrm{mag}}]) = 0.389, \qquad R^2(E_{\geq 3}+J) \approx 0.391, \qquad \Delta R^2 = 0.004 \text{ (95\% CI includes zero)}. \tag{9}$$

We do not interpret this as predictive privilege of either description. We interpret it as predictive equivalence between two descriptions that differ in the kind of information they encode: $[W_0, B_{\mathrm{mag}}]$ is a global two-scalar summary of the configuration and contains no per-site information, while $E_{\geq 3} + J$ is built from per-site counts and retains site-resolved exposure-class information.

The asymmetry between these two descriptions is not predictive—they agree as predictors of ΔW. The asymmetry is in the kind of intervention each can support. Every direction-specific generator intervention tested in this paper, including the structural $d_c = 4$-aligned arm (Section IV E), the budget-matched random-site arm (Section IV F), and the d_c-filtered random arms (Section IV F 0 a), conditions on the per-site Glauber acceptance argument $s \cdot h$. $[W_0, B_{\mathrm{mag}}]$ does not encode $s \cdot h$ at any site; therefore no per-site directional protocol can be implemented from $[W_0, B_{\mathrm{mag}}]$ alone, regardless of how well it predicts ΔW.

The dissociation is therefore between *global aggregate predictive sufficiency* and *local intervention accessibility*. It is not a claim that the $d_c = 4$ selector is the strongest causal handle in magnitude: the random-site arm achieves a larger gap (+4.70 vs +2.69 at $\varepsilon = 0.10$) and the filter decomposition cannot attribute the random-arm magnitude to any single d_c class. What the dissociation does claim is narrower and more defensible: predictive equivalence at the level of two-feature global summaries does not imply equivalence at the level of generator-level intervention. The two notions of usefulness—forecasting ΔW from a description, and supplying the per-site information that direction-specific generator interventions require—are different properties of a description and need not track each other.

H. Causal protocol in the disordered phase

The Axis 4 causal protocol was repeated at $T = 2.50$ ($N = 1500$ configurations). This is a non-trivial test: in the disordered phase, E predictively *loses* to ϕ_{var8} on ΔW (Section IV C),

so a generator bias targeted on $d_c = 4$ sites might be expected to lose its causal grip as well.

ε at $T = 2.50$	Aligned gap [95% CI]	Misaligned gap [95% CI]	Random gap [95% CI]
0.02	$+13.3\,[+7.9, +18.4]$	$-17.4\,[-22.6, -12.6]$	$+9.2\,[+3.6, +14.4]$
0.05	$+27.9\,[+22.5, +33.2]$	$-33.5\,[-38.6, -28.2]$	$+20.6\,[+15.2, +26.4]$
0.10	$+28.11\,[+23.42, +33.12]$	$-68.27\,[-73.70, -63.29]$	$+29.25$
0.20	$+27.0\,[+21.1, +32.2]$	$-136.6\,[-142.4, -131.9]$	$+38.4\,[+32.9, +43.8]$

The sign-flip discriminator passes 4/4 at $T = 2.50$, matching the ordered-phase result. The $d_c = 4$ aligned arm achieves $+28.11$ (95% CI $[+23.42, +33.12]$), an order of magnitude larger than its ordered-phase value of $+2.69$; the misaligned arm reaches -68.27 (95% CI $[-73.70, -63.29]$). Two reasons for the larger absolute gaps in the disordered phase are straightforward: there are many more $d_c = 4$ sites available to perturb, and the baseline ΔW has higher intrinsic variance in the disordered phase, leaving more room for an intervention to move it.

This result is one of the cleaner findings in the paper. Predictive advantage and direction-specific causal control behave *differently* across phases: the predictive sign reversal of Axis 2 does not propagate to the causal protocol of Axis 4. The fine description $E_{\geq 3} + J$ does not become a worse *predictor* in the disordered phase (Section IV C), and the $d_c = 4$ selector does not become a worse *causal handle* either; the only thing that reverses sign is the specific predictive comparison between the bare count E and the block-variance rival ϕ_{var8}.

I. Robustness: finite size and horizon sensitivity

a. Finite-size check. To assess finite-size stability of the Axis 2 crossover result (the primary quantitative claim), we ran $N = 500$ predictive worlds at $L \in \{32, 128\}$ at $T = 1.80$.

L	ΔR^2 (E vs ϕ_{var8})	95% CI	$\bar{E}$ sites
32	+0.156	[+0.086, +0.210]	10.6
64	+0.178	[+0.152, +0.205] (main)	42.6
128	+0.184	[+0.118, +0.248]	170.9

All three lattice sizes produce positive ΔR^2 with overlapping confidence intervals, demonstrating that the qualitative finding (fine structural counts outperform coarse block averages

in the ordered phase) is stable across $L = 32$ to $L = 128$. The mean count of exposed sites $\bar{E}$ scales roughly linearly with system area ($\bar{E} \propto L^2$), confirming that E is an extensive quantity; the relative density $\bar{E}/L^2 \approx 1\%$ is consistent across L.

b. Horizon sensitivity. We ran $N = 500$ predictive worlds at $K \in \{10, 50\}$ for $L = 64$, $T = 1.80$, using the champion descriptor $E_{\geq 3} + J$.

K (sweeps)	ΔR^2	95% CI
10 (short)	+0.369	[+0.297, +0.434]
20 (main)	+0.391	[+0.363, +0.419]
50 (long)	+0.415	[+0.349, +0.474]

The fine-over-coarse advantage is positive and substantial at all three horizons, with overlapping CIs. The advantage increases weakly with K (+0.369 to +0.415), consistent with the picture that over longer horizons more coarsening events accumulate, and cumulative interface simplification is more correlated with initial fine structural counts.

c. Cross-lattice transfer of fitted weights. Finite-size stability of the ΔR^2 statistic (above) is a different question from whether the literal fitted Ridge weights at one lattice size predict at a different lattice size. We trained the Ridge model on $L = 64$ configurations ($N = 3000$, per-site normalised features ($E_{\geq 3}/L^2$, J/L^2) and dimensionless target $\Delta W/(W_0 + \epsilon)$) and applied the fitted weights to $N = 500$ $L = 128$ configurations at the same temperature.

Evaluation	R^2
$L = 64$ in-sample (5-fold CV)	0.370
$L = 64$ weights applied at $L = 128$	0.018

The fitted weights do not transfer: R^2 falls from 0.370 to 0.018 when the same coefficients are used at $L = 128$. This is a separate observation from the ΔR^2 stability above and we do not claim cross-L weight transfer. The qualitative comparison fine $\succ$ coarse is stable across $L \in \{32, 64, 128\}$ in the ordered phase (preceding tables); the quantitative two-feature linear fit is calibrated to the lattice it was trained on.

J. Synthesis

Figure 7 summarises all four components. In the ordered phase, exposed-site and junction structure carries predictive information that coarse averaging discards; this advantage is

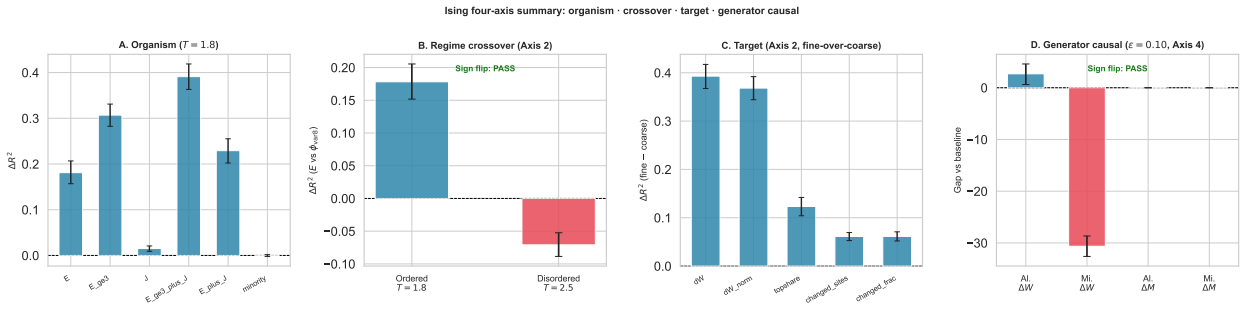

FIG. 7. **Four-component summary panel.** (**A**) Axis 1 organism comparison: $E_{\geq 3} + J$ is the champion. (**B**) Axis 2 regime crossover: fine wins ordered, coarse wins disordered. (**C**) Axis 3 target comparison: ΔW is the most fine-specific target. (**D**) Axis 4 causal test at $\varepsilon = 0.10$: aligned gap on ΔW positive, misaligned negative; ΔM comparatively small. A budget-matched random-site arm with the same directional logic produces a comparable or larger gap (Section IV F); panel D shows the structural arm only. $N = 3000$ predictive worlds, 3000 causal worlds, $L = 64$, $K = 20$.

regime-specific. Under the fine-over-coarse criterion, ΔW is the most fine-specific target. The causal sign-flip confirms that the directional conditioning is operative across phases. The aggregate-vs-local dissociation shows that global aggregate predictive sufficiency does not imply local intervention accessibility. Predictive organization, directional causal organization, and aggregate-vs-local dissociation are independently tested and jointly characterise how descriptions differ in the two senses.

V. DISCUSSION

A. What the ordered-regime fine advantage means

The fine advantage is not a trivial consequence of resolution. More bits do not automatically improve out-of-sample prediction for a specific target. What $E_{\geq 3} + J$ captures is the fragility topology of the interface: the geometrically special locations most likely to simplify in the next K steps. Coarse 8×8 block averages wash out this topology. The result is consistent with a broader principle: fine-description advantage appears when the predictive signal is concentrated in sparse, structurally special objects that coarse summaries discard—and disappears when those objects lose their sparse, special character.

B. Why the regime crossover is the main empirical test

The sign reversal between ordered and disordered phases rules out three confounds simultaneously. It rules out that fine organisms are always better because they contain more information. It rules out that the advantage is an artefact of the specific coarse rival chosen. And it provides a physical narrative tied to the phase structure: as thermal fluctuations grow near T_c, interfaces roughen, the exposed-site concept loses discriminative power [1], and bulk fluctuations become the better predictor. A finding without a crossover would be scientifically weaker.

C. Why Axis 3 requires the fine-over-coarse criterion

Measuring Axis 3 by raw R^2 would conflate fine-over-coarse advantage with raw predictability. Some targets are easy to predict in absolute terms because bulk dynamics drive them, and both fine and coarse descriptions track bulk dynamics well. The fine-over-coarse ΔR^2 criterion identifies the target where the *marginal value* of fine over coarse information is largest. That target is ΔW, consistently and for clear structural reasons. Reporting Axis 3 by raw R^2 would risk pointing to the wrong target and would weaken the structural argument.

D. Why Axis 4 is stronger than a generic gap

Any modification to the Glauber kernel differs from the unmodified kernel. What Axis 4 demonstrates is that the *direction* of the modification matters, in the way predicted by the fine structural description. The misaligned arm provides the internal negative control: reversing the alignment reverses the causal effect. This pattern supports the interpretation that the effect is direction-specific, rather than a generic perturbation effect [10, 13].

The class-level control is intentionally weaker than this directional claim: a budget-matched random-site arm with the same directional logic achieves comparable or larger gaps (Section IV F), so we do not claim that the structural arm is uniquely strong in magnitude. What both arms share, and what $[W_0, B_{\mathrm{mag}}]$ does not encode, is the per-site quantity $s \cdot h$. The Axis 4 claim is therefore about *directional conditioning at per-site resolution*, not about the privileged size of any one selection rule.

E. The dissociation and its implications

The predictor-handle dissociation (Section IV G) reveals a sharp separation between two properties of a description: predictive accuracy and causal leverage. The aggregate coarse state $[W_0, B_{\text{mag}}]$ and the fine champion $E_{\geq 3} + J$ achieve statistically indistinguishable out-of-sample R^2 for ΔW (gap CI includes zero). Yet only per-site descriptions—of which $E_{\geq 3} + J$ is one example—encode the local quantity $s \cdot h$ on which every direction-specific generator intervention we test conditions. The aggregate $[W_0, B_{\text{mag}}]$ summarises *how much* coarsening will occur on average; per-site descriptions in addition locate the configuration's directional structure ($\text{sign}(s \cdot h)$ at each site). Equal predictive power at the level of global aggregates does not imply equal accessibility for per-site directional intervention; this is the only causal-accessibility claim we make.

The class-level control (Section IV F) refines this picture: random-site targeting with the same directional logic achieves comparable or larger gaps under our protocol, and the filter decomposition (Section IV F 0 a) shows that this random-arm effect is not concentrated at any single d_c class. We therefore do not claim that $d_c = 4$ class identity is uniquely causal in magnitude; what we claim is that both the structural arm and the random arm condition on the per-site quantity $s \cdot h$, which the global aggregate $[W_0, B_{\text{mag}}]$ cannot supply. What the fine description $E_{\geq 3} + J$ provides is the ability to identify $d_c = 4$ sites and use them as specifically wall-reducing targets; the dissociation persists at the level of per-site structural vs global aggregate descriptions, not at the level of $d_c = 4$ vs any per-site operation.

F. Limitations and extensions

Results are scoped to 2D Ising, $L = 64$, $K = 20$, $T \in \{1.80, 2.50\}$ (plus near-critical probe at $T = 2.20$ and robustness checks at $L \in \{32, 128\}$ and $K \in \{10, 50\}$). Transfer to other models is not automatic. The causal result (Axis 4) is topology-sensitive: the binary-spin structure is important for a single flip to simultaneously eliminate multiple domain-wall bonds at an exposed site. Extension to Potts and Clock models would require identifying analogous geometrically special objects under multi-state spins. Extension to quantum systems (Bose-Hubbard chain) and continuous-field systems (XY, ϕ^4) requires adapting the exposed-defect concept; these are open directions.

Remaining open questions: a more complete decomposition of which d_c classes drive the random-arm gap (our filter test rules out the $d_c = 0$-only and $d_c \geq 3$-only single-class hypotheses but does not identify the active intermediate classes by direct test); how the horizon-monotone increase of the fine-over-coarse advantage extends to very long horizons ($K \gg 50$); and whether a transferable cross-lattice descriptor exists, since the literal Ridge weights at $L = 64$ do not predict well at $L = 128$ (Section IV I).

VI. CONCLUSION

We have separated two questions about descriptions in 2D kinetic Ising dynamics: how predictive accuracy depends on regime and target, and how direction-specific causal interventions depend on what a description can encode locally.

Predictive organization is regime-dependent. The fine descriptor $E_{\geq 3} + J$ leads the weak block-variance rival ϕ_{var8} by $\Delta R^2 = +0.391$ (95% CI [+0.363, +0.419]) at $T = 1.80$, ties the stronger global aggregate $[W_0, B_{\text{mag}}]$ ($\Delta R^2 = 0.004$, CI includes zero), and the simpler E undergoes a smooth temperature-dependent sign reversal: $+0.178 \to +0.108 \to -0.005 \to -0.001 \to -0.071$ across $T \in \{1.80, 2.00, 2.20, 2.35, 2.50\}$. Robustness checks across $L \in \{32, 64, 128\}$ and $K \in \{10, 20, 50\}$ confirm stability of the ΔR^2 statistic, although the fitted Ridge weights themselves do not transfer across lattice size.

Direction-specific causal organization is robust across that same temperature range. The aligned/misaligned generator protocol passes the sign-flip discriminator 4/4 at $T = 1.80$ and 4/4 at $T = 2.50$. The temperature reversal of the predictive comparison does not propagate to the causal protocol; the same per-site directional intervention works in both phases.

Causal magnitude is not single-selector. A budget-matched random-site arm with the same directional logic produces gaps at least as large as the structural $d_c = 4$ arm (+4.70 vs +2.69 at $\varepsilon = 0.10$), and a filter decomposition cannot attribute the random-arm magnitude to any single d_c class. We therefore do not claim that the structural selector is uniquely strongest in causal magnitude; the surviving causal regularity is sign structure, not selector-magnitude privilege.

The surviving dissociation is aggregate-vs-local. $[W_0, B_{\text{mag}}]$ matches $E_{\geq 3} + J$ as a predictor of ΔW but does not encode the per-site quantity $s \cdot h$ on which every direction-specific generator intervention conditions. Predictive equivalence at the level of global summaries

does not imply equivalence at the level of generator-level intervention.

The robust regularities are therefore: predictive structure that depends on regime and varies smoothly with temperature; direction-specific causal sign control that does not inherit the predictive sign reversal; and a dissociation that lives at the boundary between global aggregate prediction and local intervention, not at the boundary between one preferred selector and the rest of the descriptor space.

ACKNOWLEDGMENTS

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DATA AND CODE AVAILABILITY

All simulation code, analysis scripts, and generated data are available at https://github.com/kunalb541/Ising.

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[1] A. J. Bray, Theory of phase-ordering kinetics, Adv. Phys. **43**, 357 (1994).

[2] B. Derrida, A. J. Bray, and C. Godrèche, Non-trivial exponents in the zero temperature dynamics of the 1D Ising and Potts models, J. Phys. A **27**, L357 (1994); B. Derrida, V. Hakim, and V. Pasquier, Exact exponent for the number of persistent spins in the zero-temperature dynamics of the one-dimensional Potts model, Phys. Rev. Lett. **75**, 751 (1995).

[3] I. M. Lifshitz, Kinetics of ordering during second-order phase transitions, Zh. Eksp. Teor. Fiz. **42**, 1354 (1962) [Sov. Phys. JETP **15**, 939 (1962)]; C. Wagner, Theorie der Alterung von Niederschlägen durch Umlösen (Ostwald-Reifung), Z. Elektrochem. **65**, 581 (1961).

[4] R. J. Glauber, Time-dependent statistics of the Ising model, J. Math. Phys. **4**, 294 (1963).

[5] E. Hoel, L. Albantakis, and G. Tononi, Quantifying causal emergence shows that macro can beat micro, Proc. Natl. Acad. Sci. USA **110**, 19790 (2013).

[6] E. Hoel, L. Albantakis, W. Marshall, and G. Tononi, Can the macro beat the micro?, Neurosci. Conscious. **2016**, niw012 (2017).

[7] E. Ising, Beitrag zur Theorie des Ferromagnetismus, Z. Phys. **31**, 253 (1925).

[8] N. Israeli and N. Goldenfeld, Coarse-graining of cellular automata, emergence, and the predictability of complex systems, Phys. Rev. E **73**, 026203 (2006).

[9] L. P. Kadanoff, Scaling laws for Ising models near T_c, Physics (Long Island City, N.Y.) **2**, 263 (1966).

[10] J. Pearl, *Causality: Models, Reasoning, and Inference*, 2nd ed. (Cambridge University Press, Cambridge, 2009).

[11] V. Spirin, P. L. Krapivsky, and S. Redner, Fate of zero-temperature Ising ferromagnets, Phys. Rev. E **63**, 036118 (2001).

[12] K. G. Wilson, The renormalization group: critical phenomena and the Kondo problem, Rev. Mod. Phys. **47**, 773 (1975).

[13] J. Woodward, *Making Things Happen: A Theory of Causal Explanation* (Oxford University Press, Oxford, 2003).

[14] F. Noé, S. Olsson, J. Köhler, and H. Wu, Boltzmann generators: sampling equilibrium states of many-body systems with deep learning, Science **365**, eaaw1147 (2019).

[15] J. Wang, S. Olsson, C. Wehmeyer, A. Pérez, N. E. Charron, G. de Fabritiis, F. Noé, and C. Clementi, Machine learning of coarse-grained molecular dynamics force fields, ACS Cent. Sci. **5**, 755 (2019).